Some IMBHs are thought to form in the cores of dense clusters of stars called globular clusters, and some of them are likely to find their way into the nuclei of galaxies, where gigantic black holes reside.

An example is Andromeda, the nearest large galaxy to our own (Figure 7.4), in whose nucleus lurks a Gargantua-sized black hole: 100 million solar masses. Huge numbers of stars are drawn into the vicinity of such gigantic black holes; as many as a thousand stars per cubic light-year. When an IMBH passes through such a dense region, it gravitationally deflects the stars, creating a wake with enhanced density behind itself (Figure 7.4). The wake pulls on the IMBH gravitationally, slowing the IMBH down, a process called “dynamical friction.” As the IMBH very gradually slows, it sinks deeper into the vicinity of the gigantic black hole. In this manner, Nature could provide Cooper, in my interpretation of *Interstellar*, with the IMBHs that he needs for his slingshots.[19]

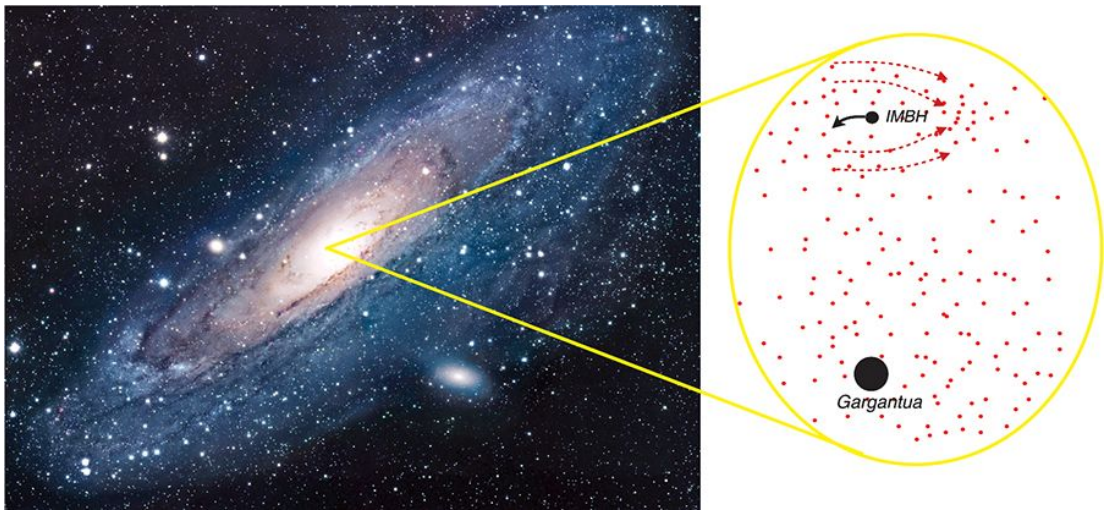


Fig. 7.4. *Left*: The Andromeda galaxy, which harbors a Gargantua-sized black hole. *Right*: The dynamical friction by which an IMBH will gradually slow down and sink into the vicinity of the gigantic black hole.